

Court Case Management System (CCMS)

Agile User Stories & Engineering Acceptance Criteria

Target Frontend: Angular 21

Target Backend: .NET 10 Web API

Infrastructure: Azure AKS / SQL 2022

Project Overview & Scope:

CCMS is a distributed enterprise web application designed to fully digitize the manual workflow of courts issuing legal orders (Account Freezes and Balance Enquiries) to banking organizations. The system features multi-role portals (Court Officers and Bank Officers), strict PII data masking, automated batch processing with hierarchical cascade resolution logic, and a containerized cloud architecture.

Epic 1: Authentication & Shared Security Framework

US-1.1 Unified Multi-Role Authentication Gateway

As a registered Court Officer or Bank Officer,
I want to authenticate into the system using a single, unified login interface,
So that I am automatically routed to my appropriate, role-specific operational dashboard.

ACCEPTANCE CRITERIA

- A single web login interface serves both system roles seamlessly.
- The authentication system must be fully stateless and governed by JWT Bearer tokens.
- The JWT signing secret must be a strong random string (minimum 32 characters) loaded dynamically via environment variables (Kubernetes Secrets / Azure Key Vault).
- Upon successful token validation, the application layer must decipher claims and explicitly route users: Court Officers to the Court Dashboard, Bank Officers to the Bank Dashboard.
- There is no self-registration framework; the system database seed file must hard-code two initial operational profiles for demonstration.

US-1.2 Automated API-Level PII Data Masking

As a System Compliance Auditor,

I want highly sensitive defendant identification numbers and financial parameters to be automatically obfuscated before transmitting,

So that Personally Identifiable Information (PII) is securely protected at rest in transit, and restricted on-screen.

ACCEPTANCE CRITERIA

- Data masking must be actively applied to three targeted data fields: Defendant Aadhaar Numbers, PAN Numbers, and Core Bank Account Numbers.
- The masking logic must execute strictly within the API response serialization layer before data reaches the network or Angular application layer.
- The masking presentation rule is fixed: only the last four sequential characters can be readable text (e.g., *****1234).
- Raw values must remain completely intact inside the SQL Server database to support automatic server-side batch matching tasks.

Epic 2: Module 1 — Digital Court Portal Workspace

US-2.1 Court Officer Summary Dashboard

As a Court Officer,

I want to view a centralized metric dashboard tracking all legal cases I have filed,

So that I can understand processing backlogs and trace workflow completions in real-time.

ACCEPTANCE CRITERIA

- The dashboard must group and aggregate the total volume of case records raised by the logged-in officer according to their active lifecycle states.
- The UI must display rapid numeric data visualization summaries for: Pending, Account Validated, Account Not Found, Freeze Applied, and Balance Provided states.
- Actionable hotlinks must enable instantaneous navigation to create a new legal case file or view detailed summaries of existing historical logs.

US-2.2 Legal Order Formulation & Multi-Document Upload

As a Court Officer,

I want to digitally register a new case file containing comprehensive complainant, defendant, and multi-file evidence structures,

So that physical paperwork is eliminated and legally binding orders are channeled automatically to banking networks.

ACCEPTANCE CRITERIA

- The Angular system must present a robust Reactive Form validating: Complainant Metadata, Defendant Identity (Full Name, Aadhaar, PAN, Target Bank, Target Account Number), and Core Order Classification (Freeze Account or Balance Enquiry).
- If the selection matches Freeze Account, the application must demand a strict numeric constraint entry outlining the requested freeze limit volume.
- The submission payload requires three distinct document attachments: Signed Court Order PDF, Defendant Aadhaar Copy, and Defendant PAN Copy.
- The file attachment engine must strictly enforce data whitelist policies: accepting only PDF, JPG, or PNG extensions, with a maximum payload cap of 5 MB per individual file.
- The core API backend must automatically generate a globally unique, sequential tracking key during database ingestion using the exact format schema: CCMS-YYYYMMDD-XXXX.
- Newly registered records must default into an initial workflow state of Pending.

US-2.3 Unified Court Read-Only Case Tracker

As a Court Officer,

I want to open a detailed summary of a specific case submission,

So that I can securely view the matching outcome status and formal banking responses in a read-only environment.

ACCEPTANCE CRITERIA

- The UI view must lock all inputs to a completely read-only status.
- Sensitive identification criteria (Aadhaar, PAN, Account Numbers) must render using the API masking format.
- The tracker must explicitly show current operational status, system matching logs, and, if completed, the verified banking metrics (Balance Provided, Applied Freeze Limits, Bank Officer Remarks, and precise timestamps).

Epic 3: Module 2 — Automated Account Validation Engine

US-3.1 Scheduled Hierarchical Cascade Matching Job

As an Automated System Background Service,

I want to wake up on a rigorous 15-minute schedule to inspect unverified cases,

So that I can automatically resolve customer matching validation queries without consuming human resource hours.

ACCEPTANCE CRITERIA

- The process must execute as an isolated, asynchronous .NET BackgroundService hosted engine.
- Every distinct execution run must narrow its processing scope exclusively to cases residing in a Pending state. Any data record already marked as resolved or intermediate must remain entirely unmodified.
- The account search algorithm must follow a cascade hierarchy model against bank data tables: Match Attempt 1: Core Account Number string. If no match is found, switch to Match Attempt 2: Aadhaar string. If no match is found, switch to Match Attempt 3: PAN string.
- **Successful Search Event:** Change case lifecycle state directly to Account Validated and instantly bind target information (current account balance, active flags) to the case table.
- **Failed Search Event:** Automatically short-circuit the record lifecycle directly to the terminal state Account Not Found and output an automated system statement: "No matching account found in bank records."
- The background runner must persist analytics telemetry records after every execution cycle (aggregating total records verified, error thresholds, and duration metrics).

Epic 4: Module 3 — Operational Bank Portal Workspace

US-4.1 Bank Management Console & Manual Overrides

As a Bank Officer,

I want to view a broad overview of incoming legal caseload volume and control execution layers,

So that I can monitor operational bottlenecks and manually trigger validation pipelines for immediate legal enforcement.

ACCEPTANCE CRITERIA

- The user interface must organize records by business priority metrics: cases demanding manual confirmation action, items actively idling in the validation worker queue, and historically closed files.
- The dashboard must render the exact timestamp indicating when the automated validation engine last successfully finished a data sweep.
- An explicit, secure management button must be available to let the Bank Officer manually trigger the background service immediately without waiting for the 15-minute cron scheduler.

US-4.2 Tabbed Action Inbox Management

As a Bank Officer,

I want to utilize an organized, tab-segregated workspace container,

So that I can safely partition my daily operational tasks from system-completed files.

ACCEPTANCE CRITERIA

- **Tab 1 (Awaiting Action):** Populated solely with records marked Account Validated.
- **Tab 2 (Completed):** Displays records processed by an officer to terminal conditions (Freeze Applied / Balance Provided).
- **Tab 3 (Auto-Resolved):** Houses read-only logs closed via the automated system worker (Account Not Found).
- **Tab 4 (Pending Batch):** Contains informational items currently in a Pending queue.

US-4.3 Legal Order Actioning & Response Form Idempotency

As a Bank Officer,

I want to evaluate court validation details, inspect digital attachments, and lodge official compliance confirmation payloads,

So that court directives are legally executed and final outcomes are reflected on the Court Portal.

ACCEPTANCE CRITERIA

- The workflow interface must provide simple download mechanisms to view the original uploaded court files at any time.
- **Scenario A — Balance Enquiry Directive:** The application must pre-fill the input field with the financial balance pulled during the automated verification run, allowing the officer to make manual updates and append text remarks before final submission. Changes status to Balance Provided.
- **Scenario B — Freeze Account Directive:** The form must intentionally render an empty, unpopulated freeze input box. The officer must open the uploaded PDF order, manually identify the freeze amount, fill out the form, and append verification remarks. Changes status to Freeze Applied.
- **Strict Idempotency Guard:** The API logic and user interface must aggressively protect the integrity of a terminal case. Once a response model is successfully submitted, the active editing component must instantly vanish. The API router must throw a strict exception block rejecting any secondary modification attempts for that record.